

Lab 01 Propositional Logic

Objective

Solving exercises from the textbook in chapter 1.1

Current Lab Learning Outcomes (LLO)

By completion of the lab, the students should be able to:

1. understand propositional logic concepts and practice the truth tables.

Lab Requirements

Students allowed using their lecture notes in the lab in order to solve the exercises.

Lab Assessment

- 1- Divide students to groups and let them to solve the given example.
- 2- Discuss the answers with the groups and write on board the optimal solution.

Lab Description

1. Which of these sentences are propositions? What are the truth values of those that are propositions?
 - a) Riyadh is the capital of Kuwait.
 - b) Cairo is the capital of Egypt
 - c) $2 + 3 = 5$.
 - d) $5 + 7 = 10$.
 - e) $x + 2 = 11$.
 - f) Answer this question.
2. What is the negation of each of these propositions?
 - a) Mei has an MP3 player.
 - b) There is no pollution in New Jersey.
 - c) $2 + 1 = 3$.
3. Let p and q be the propositions p : It is below freezing. q : It is snowing. Write these propositions using p and q and logical connectives(including negations).
 - a. It is below freezing and snowing.
 - b. It is below freezing but not snowing.
 - c. It is not below freezing and it is not snowing.
 - d. It is either snowing or below freezing (or both).
 - e. If it is below freezing, it is also snowing.
 - f. Either it is below freezing or it is snowing, but it is not snowing if it is below freezing.
 - g. That it is below freezing is necessary and sufficient for it to be snowing.

4. Let p and q be the propositions “Swimming at the New Jersey shore is allowed” and “Sharks have been spotted near the shore,” respectively. Express each of these compound propositions as an English sentence.
- a) $\neg q$ b) $p \wedge q$ c) $\neg p \vee q$ d) $p \rightarrow \neg q$ e) $\neg q \rightarrow p$ f) $\neg p \rightarrow \neg q$ g) $p \leftrightarrow \neg q$
h) $\neg p \wedge (p \vee \neg q)$.
5. How many rows appear in a truth table for each of these compound propositions?
- a. $(q \rightarrow \neg p) \vee (\neg p \rightarrow \neg q)$
b. $(p \vee \neg t) \wedge (p \vee \neg s)$
c. $(p \rightarrow r) \vee (\neg s \rightarrow \neg t) \vee (\neg u \rightarrow v)$
d. $(p \wedge r \wedge s) \vee (q \wedge t) \vee (r \wedge \neg t)$
6. Construct a truth table for each of these compound propositions.
- a) $(p \vee \neg q) \rightarrow q$ b) $(p \vee q) \rightarrow (p \wedge q)$ c) $(p \vee q) \rightarrow (p \oplus q)$ d) $(p \vee q) \oplus (p \wedge q)$
e) $(p \leftrightarrow q) \oplus (\neg p \leftrightarrow \neg r)$ f) $(p \oplus q) \rightarrow (p \oplus \neg q)$